

CP020003 — Artificial Intelligence 2026

Khon Kaen University

Week 1 — Homework Assignment

Objective

This week, you will practice fundamental data engineering and Exploratory Data Analysis (EDA) techniques using pandas on the Heart Disease (KKU Version) dataset. Your goal is to answer 10 meaningful questions that extract insights from the data using Python — progressing from basic summary statistics up to a full preprocessing pipeline ready for machine learning models.

Dataset Description

The Heart Disease (KKU Version) dataset contains clinical measurements collected from patients undergoing cardiac evaluation. Each row represents one patient.

Column	Description
age	Age of the patient (years)
sex	Biological sex: 1 = male, 0 = female
cp	Chest pain type: 0 = typical angina, 1 = atypical angina, 2 = non-anginal, 3 = asymptomatic
trestbps	Resting blood pressure (mm Hg) on admission
chol	Serum cholesterol (mg/dL)
fbs	Fasting blood sugar > 120 mg/dL: 1 = true, 0 = false
restecg	Resting ECG results: 0 = normal, 1 = ST-T wave abnormality, 2 = left ventricular hypertrophy
thalach	Maximum heart rate achieved during stress test
exang	Exercise-induced angina: 1 = yes, 0 = no
oldpeak	ST depression induced by exercise relative to rest
slope	Slope of peak exercise ST segment: 0 = upsloping, 1 = flat, 2 = downsloping
ca	Number of major vessels (0–3) colored by fluoroscopy
thal	Thalassemia: 0 = normal, 1 = fixed defect, 2 = reversible defect
target	Heart disease diagnosis: 1 = disease present, 0 = no disease



Assignment Instructions

You must:

1. Load the dataset into a pandas DataFrame using the GitHub URL provided below.
2. Answer all 10 questions in order. Each question includes a clear description of what output is expected.
3. Write clean, well-commented Python code for each question.
4. Use pandas and NumPy only — no scikit-learn until the Bonus section.
5. Show your outputs clearly (printed tables, computed values, etc.).
6. (Bonus) Apply a scikit-learn classifier (e.g., RandomForestClassifier) to the preprocessed data from Q10 and report accuracy, precision, and recall.



Questions — Easy to Hard

#	Question	Level	Points
Q1	Dataset overview: total patients, disease vs no-disease count and percentage	Easy	★☆☆☆☆
Q2	Age distribution: minimum, maximum, mean, and median age	Easy	★☆☆☆☆
Q3	Gender distribution: patient counts and disease rate per gender	Easy	★☆☆☆☆
Q4	Cholesterol outliers: list patients with chol > 300 mg/dL	Easy	★☆☆☆☆
Q5	Age group analysis: create age group column; compute disease rate per group	Medium	★★★★☆
Q6	Chest pain type vs disease: disease rate sorted by chest pain type	Medium	★★★★☆
Q7	Pearson correlation: rank all features by correlation with target	Medium	★★★★☆
Q8	Fasting blood sugar effect: % with high FBS; compare disease rates	Medium	★★★★☆
Q9	Multi-feature risk profile: mean of age, trestbps, chol, thalach, oldpeak by sex × target	Hard	★★★★☆
Q10	Full preprocessing pipeline: fill NaN, verify numeric encoding, min-max scale, validate	Hard	★★★★★

Getting Started — Dataset Code Snippet

Load the dataset directly from GitHub with the following Python code:

```
import pandas as pd
import numpy as np

DATASET_URL = (
    "https://raw.githubusercontent.com/"
    "kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1/"
    "main/dataset/heart_kku_version.csv"
)

df = pd.read_csv(DATASET_URL)
print(f"Shape: {df.shape}")
df.head()
```

Tip You can also open this notebook directly in Google Colab — no local setup required!

Deliverables

Submit one of the following:

- A Jupyter Notebook (.ipynb) — preferred
- A Google Colab link (make sure sharing is set to Anyone with the link)
- A Python script (.py) with clear section comments

Your submission must include:

- Code that solves all 10 questions
- Comments explaining your logic for each question
- Clear printed outputs for every question
- (Bonus) An extra clearly labeled section: "Bonus: scikit-learn Classifier"

Evaluation Criteria

Criterion	Weight
Correctness and completeness of all 10 answers	40%
Appropriate use of pandas (efficient, idiomatic code)	25%
Code clarity and quality of comments	20%
Insight and interpretation of results	15%

References & Acknowledgements

This assignment was designed with gratitude to the open-source communities that make AI education accessible to everyone:

- Kaggle — dataset hosting and community notebooks
- PyTorch — deep learning framework and documentation
- Python & Pandas — the foundation of modern data science

Submission Deadline

The deadline for this assignment will be announced and agreed upon during class. Please make sure to take note of the finalized date as discussed in our session.

Late submissions may be penalised unless prior arrangements have been made with the instructor.